

Star Vetter - confirmed eclipsing binary candidates

TIC	Period (d)	Depth	BLS SNR	Pixel SNR	Offset	Tier
437150434	5.2853	15.003%	1081	85.33	1"	Clean
237115819	14.7921	8.883%	507	27.82	4"	Clean
66732476	3.1087	9.814%	370	26.93	8"	Clean
29495157	5.1474	8.579%	333	52.62	5"	Alias
119459681	5.9442	15.977%	311	68.15	2"	Clean
411434067	1.3536	4.289%	193	33.70	4"	Alias
330901906	3.7321	0.498%	60	87.24	4"	Thin margin
303470667	5.4319	2.220%	57	5.02	9"	Marginal
273158289	1.1519	1.255%	41	17.24	2"	Thin margin
111149941	5.2401	0.045%	10	9.73	21"	Ambiguous
251898081	27.3089	18.630%	5535	464.16	3"	Alias
467214801	0.7609	0.950%	2134	21.36	0"	Alias
399770041	1.5802	4.291%	489	36.92	1"	Clean
349701853	30.3427	28.445%	343	51.05	6"	Clean
233459038	25.3384	8.315%	237	84.46	5"	Alias
281310348	1.7702	7.273%	180	57.87	2"	Clean
318887646	14.2817	3.605%	150	6.94	13"	Clean
444465823	18.5133	3.476%	147	25.18	1"	Alias
380943251	0.7209	7.667%	101	15.26	3"	Clean
66716482	8.0283	3.329%	65	21.51	7"	Clean
309656476	28.9204	0.050%	25	13.80	14"	Clean
229494833	15.3839	0.037%	15	5.19	12"	Alias
44533527	14.5952	0.101%	12	7.33	16"	Clean
220423668	3.6511	4.525%	755	17.67	6"	Alias
365998904	1.2494	2.398%	375	15.52	3"	Clean
233452636	1.9063	5.467%	202	8.94	2"	Thin margin
437171483	1.4735	4.947%	923	73.37	3"	Clean
411457490	2.4324	5.531%	431	58.00	1"	Alias
295373509	5.4219	8.026%	247	22.12	2"	Clean
374158907	14.6563	4.974%	179	21.40	2"	Alias

TIC 365764494

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 0.780297 d (ALIASED, see below)
-> corrected period: 1.560595 d
Epoch (t0) 2825.3446 BTJD
Duration 0.0840 d
Depth 0.778 % (7,782 ppm)
SNR 1042.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

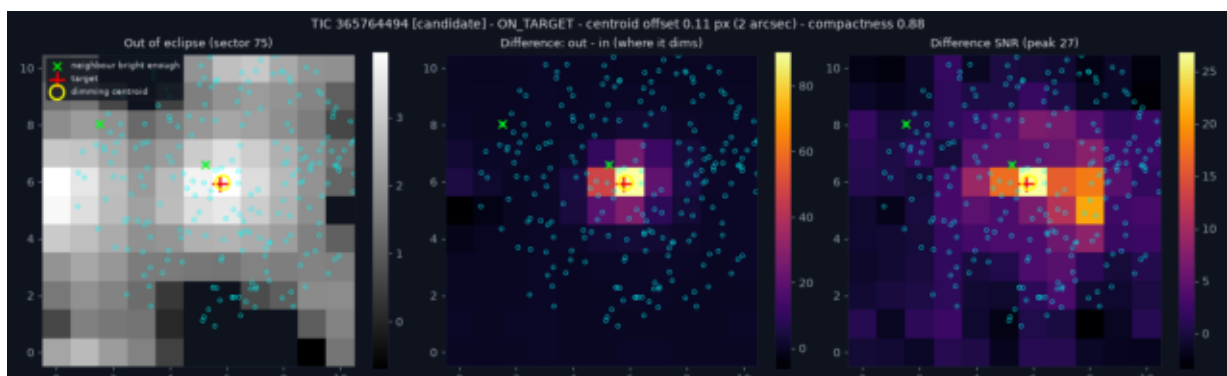
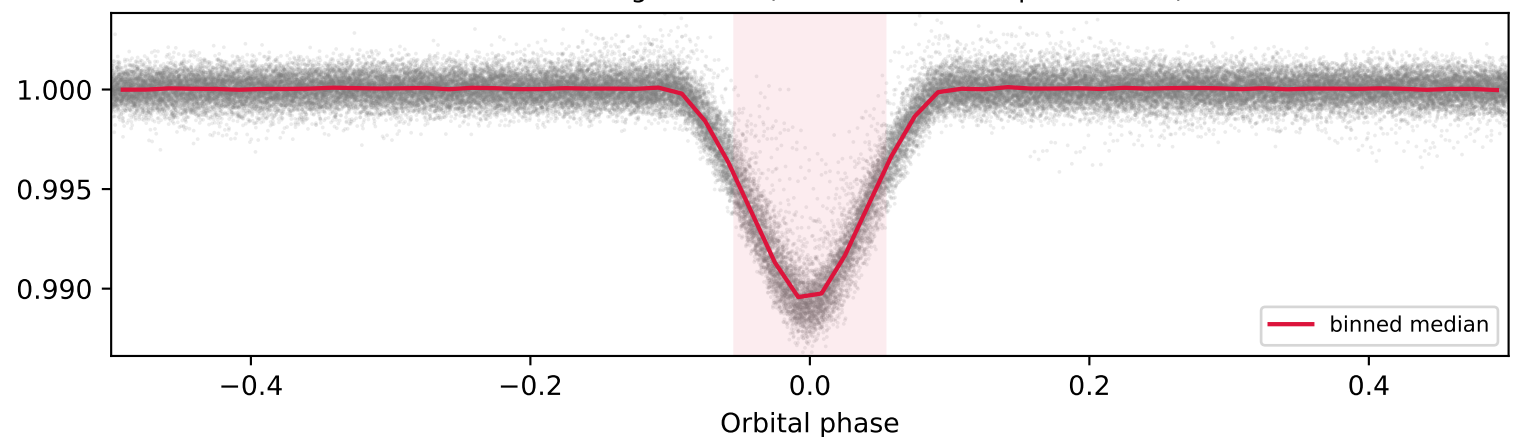
Eclipses observed 166 (gate ≥ 3)
Robust depth ratio 1.04 (gate ≥ 0.50)
Robust depth sigma 737.0 (gate ≥ 7.0)
Odd/even z-score 41.92 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 75 light-curve dip in this sector: 354.6 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 26.91 (gate ≥ 5.0) Compactness 0.88 (gate ≥ 0.50)
Dimming centroid offset 0.11 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 365764504
0.89 px away, dTmag +3.94 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 41.92 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 1.560595 d, in any write-up.

TIC 399544063

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.692904 d
Epoch (t0) 2882.3628 BTJD
Duration 0.0400 d
Depth 3.130 % (31,295 ppm)
SNR 947.4

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 66 (gate ≥ 3)
Robust depth ratio 1.26 (gate ≥ 0.50)
Robust depth sigma 58.0 (gate ≥ 7.0)
Odd/even z-score 1.82 (consistent)
Secondary eclipse not detected

CATALOG CROSS-MATCH (novelty)

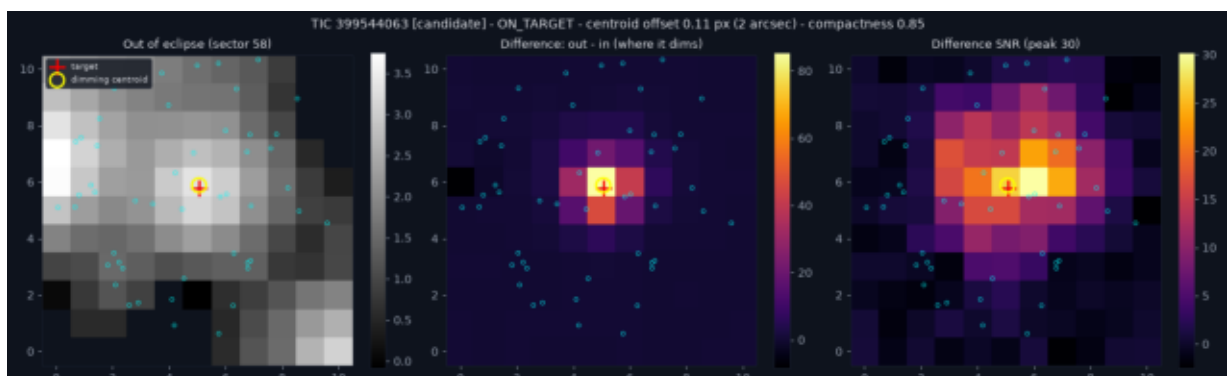
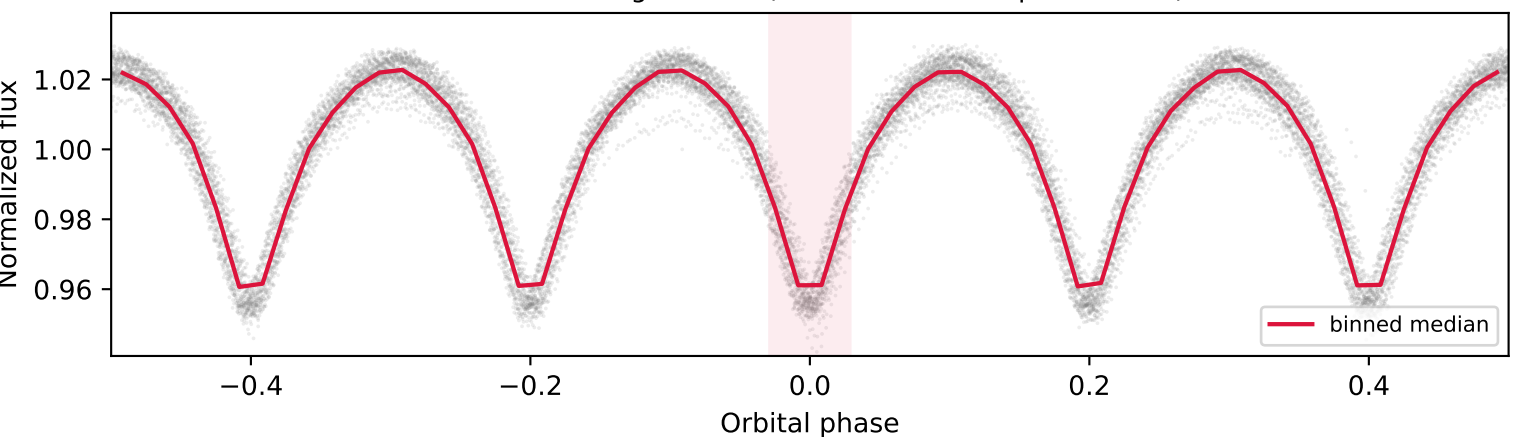
VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 58 light-curve dip in this sector: 36.7 sigma t0 refined by -0.3975 phase
Difference-image peak SNR 30.25 (gate ≥ 5.0) Compactness 0.85 (gate ≥ 0.50)
Dimming centroid offset 0.11 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 229146166

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 1.995129 d (ALIASED, see below)
-> corrected period: 3.990257 d
Epoch (t0) 1354.6820 BTJD
Duration 0.0580 d
Depth 15.426 % (154,264 ppm)
SNR 446.3

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

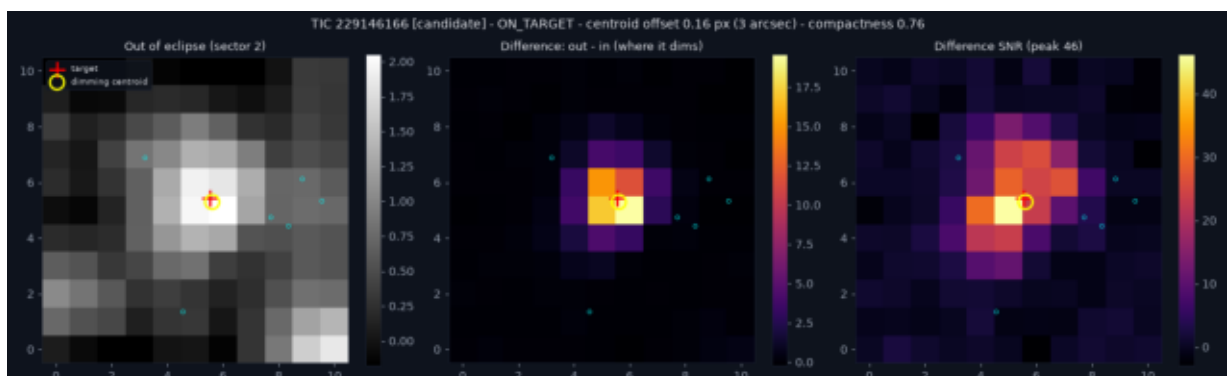
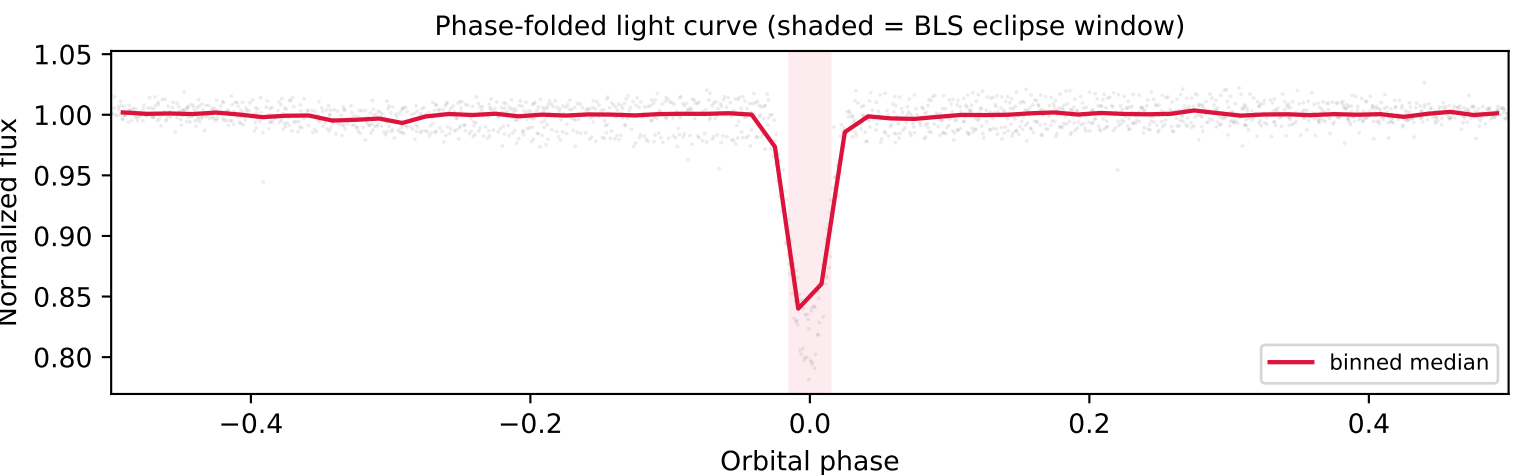
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 13 (gate ≥ 3)
Robust depth ratio 1.03 (gate ≥ 0.50)
Robust depth sigma 103.1 (gate ≥ 7.0)
Odd/even z-score 10.40 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 2 light-curve dip in this sector: 81.2 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 46.20 (gate ≥ 5.0) Compactness 0.76 (gate ≥ 0.50)
Dimming centroid offset 0.16 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 10.40 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 3.990257 d, in any write-up.

TIC 238618726

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.166692 d
Epoch (t0) 1494.5954 BTJD
Duration 0.1700 d
Depth 11.824 % (118,235 ppm)
SNR 374.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

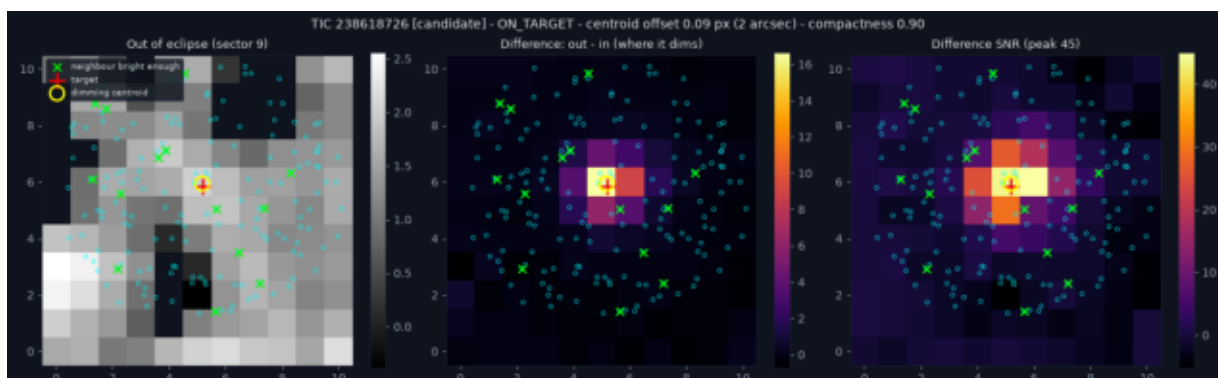
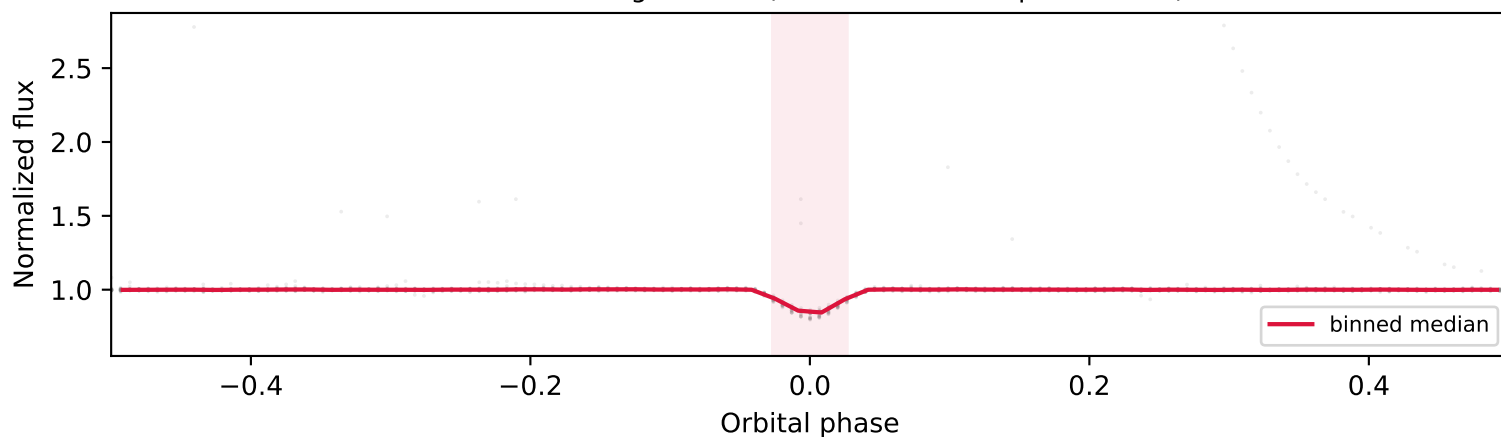
Eclipses observed 19 (gate ≥ 3)
Robust depth ratio 1.02 (gate ≥ 0.50)
Robust depth sigma 259.3 (gate ≥ 7.0)
Odd/even z-score 0.28 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 9 light-curve dip in this sector: 176.5 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 45.08 (gate ≥ 5.0) Compactness 0.90 (gate ≥ 0.50)
Dimming centroid offset 0.09 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 238618737
1.03 px away, dTmag +0.25 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



TIC 216935767

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	0.985930 d	VSX	no match
Epoch (t0)	1625.4998 BTJD	ASAS-SN	no match
Duration	0.1200 d	Gaia eclipsing-binary table	no match
Depth	3.000 % (29,997 ppm)	Gaia variable classification	no match
SNR	281.9	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	21 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.01 (gate >= 0.50)		
Robust depth sigma	60.5 (gate >= 7.0)		
Odd/even z-score	1.10 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

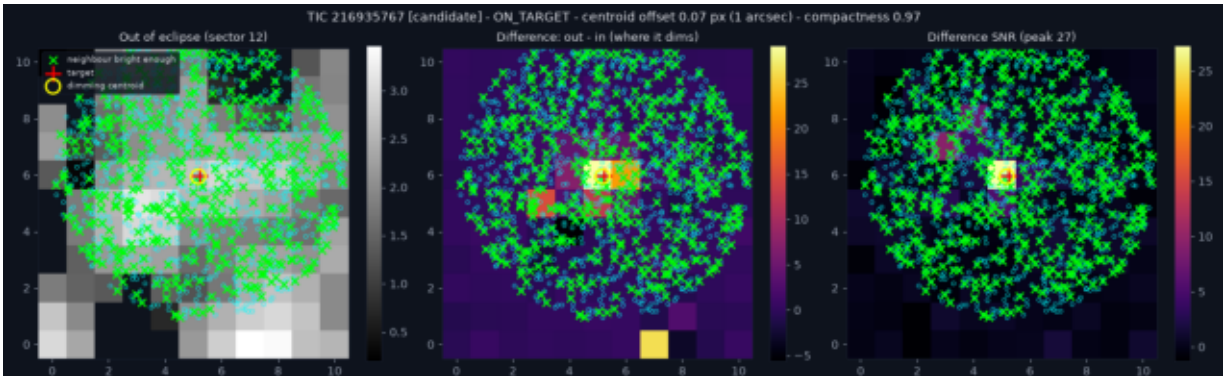
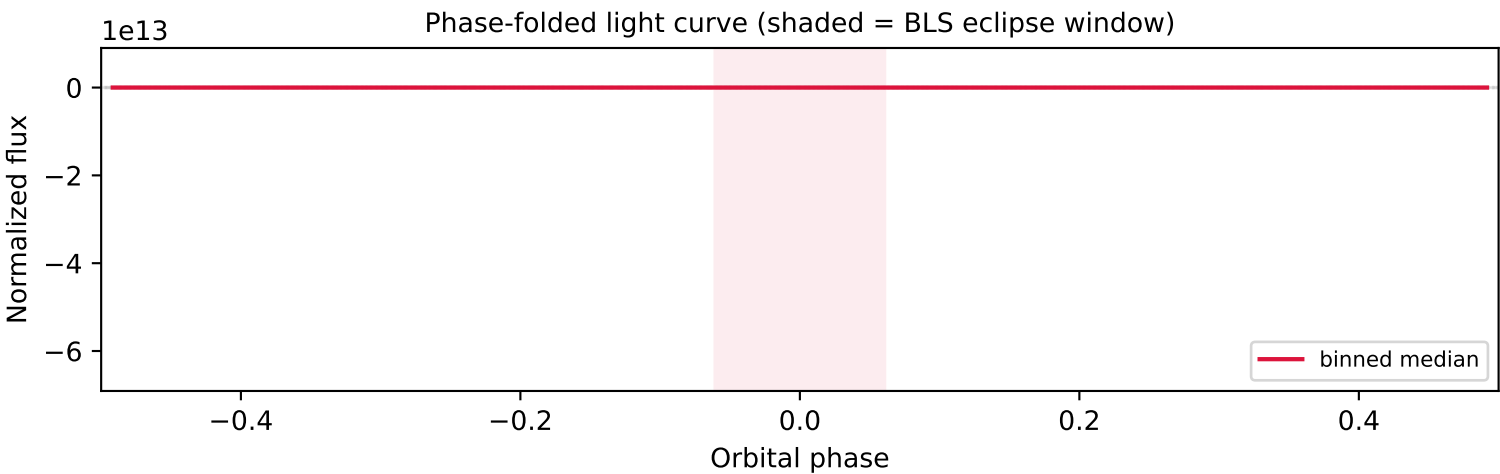
Sector 12 light-curve dip in this sector: 60.6 sigma t0 refined by 0.0150 phase

Difference-image peak SNR 27.34 (gate >= 5.0) Compactness 0.97 (gate >= 0.50)

Dimming centroid offset 0.07 px (1 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 216935780

0.21 px away, dTmag -1.42 - farther from the dimming than the target



TIC 333145823

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.898458 d
Epoch (t0) 1468.9869 BTJD
Duration 0.0400 d
Depth 1.183 % (11,830 ppm)
SNR 270.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

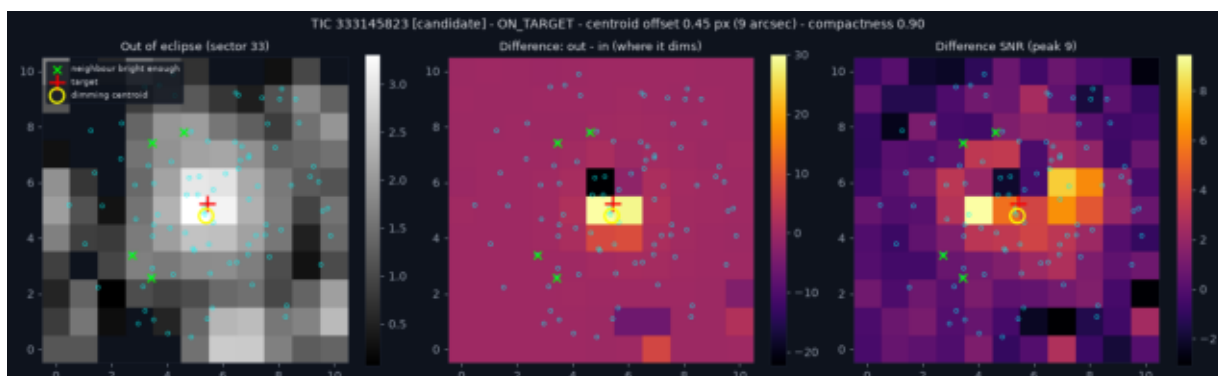
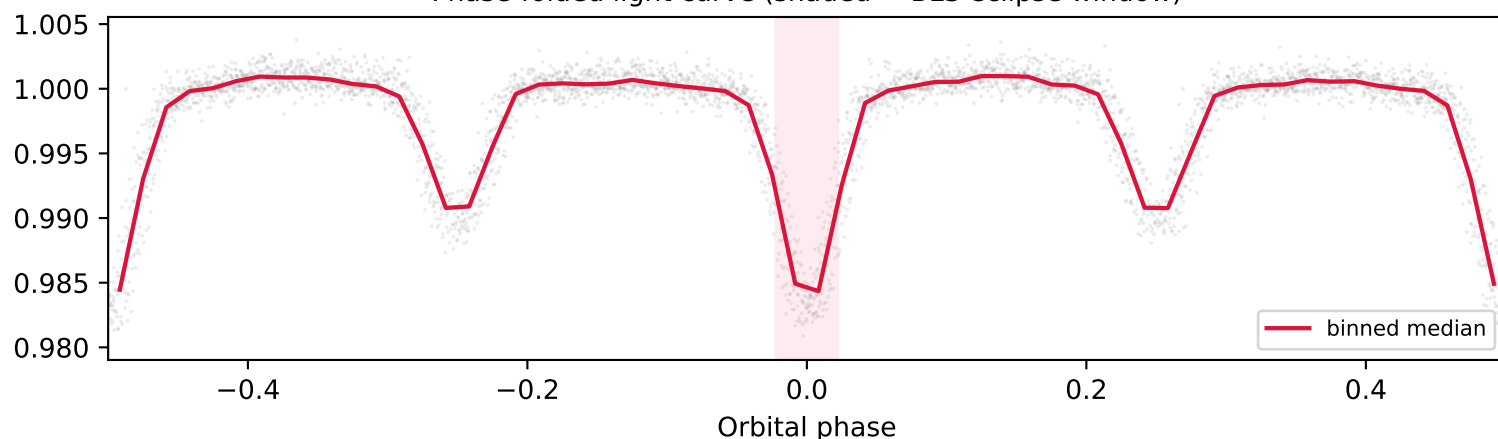
Eclipses observed 28 (gate ≥ 3)
Robust depth ratio 1.18 (gate ≥ 0.50)
Robust depth sigma 113.2 (gate ≥ 7.0)
Odd/even z-score 0.49 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 33 light-curve dip in this sector: 70.5 sigma t0 refined by -0.5000 phase
Difference-image peak SNR 9.47 (gate ≥ 5.0) Compactness 0.90 (gate ≥ 0.50)
Dimming centroid offset 0.45 px (9 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 333145798
2.96 px away, dTmag +4.67 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 459533211

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 2.065815 d
Epoch (t0) 1570.7513 BTJD
Duration 0.1200 d
Depth 8.842 % (88,416 ppm)
SNR 185.4

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 13 (gate ≥ 3)
Robust depth ratio 0.99 (gate ≥ 0.50)
Robust depth sigma 105.7 (gate ≥ 7.0)
Odd/even z-score 0.15 (consistent)
Secondary eclipse not detected

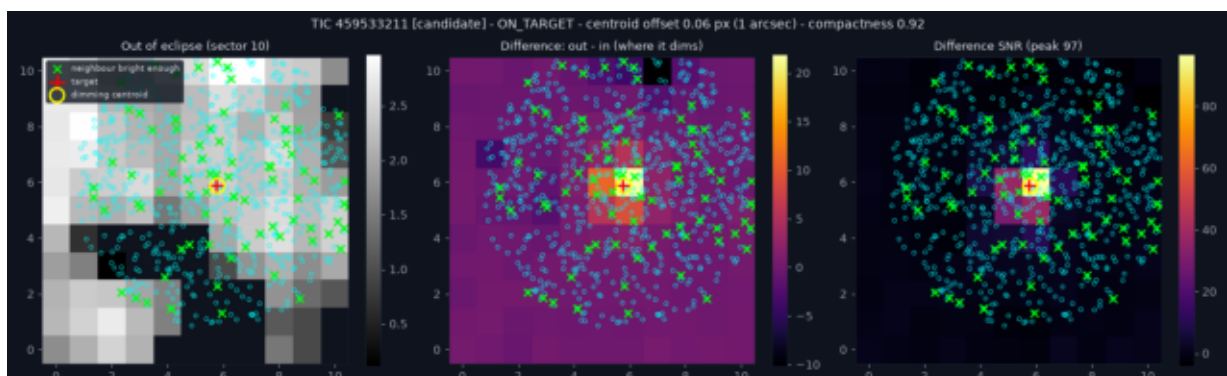
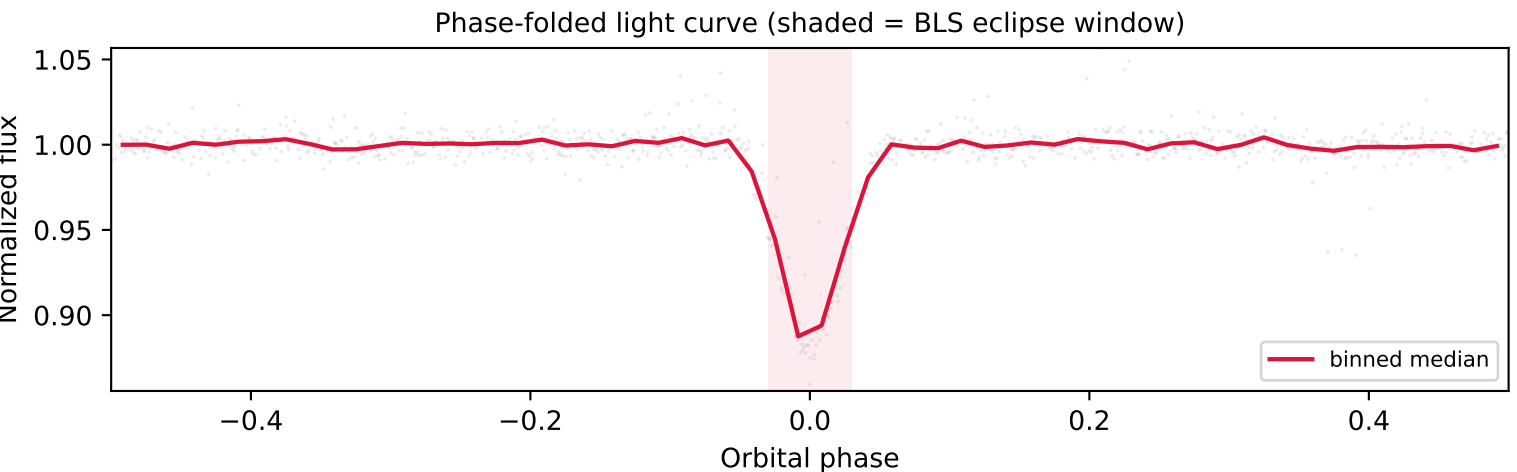
CATALOG CROSS-MATCH (novelty)

VSX no match
ASAS-SN no match
Gaia eclipsing-binary table no match
Gaia variable classification no match
TESS EB catalog (Prsa+2022) no match
ExoFOP TOI list no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 10 light-curve dip in this sector: 103.4 sigma t0 refined by -0.0075 phase
Difference-image peak SNR 96.89 (gate ≥ 5.0) Compactness 0.92 (gate ≥ 0.50)
Dimming centroid offset 0.06 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 459533193
0.40 px away, dTmag +1.96 - farther from the dimming than the target



No caveats - clean detection on every check.

TIC 306294675

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 2.982219 d
Epoch (t0) 1570.3514 BTJD
Duration 0.1200 d
Depth 2.434 % (24,341 ppm)
SNR 179.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

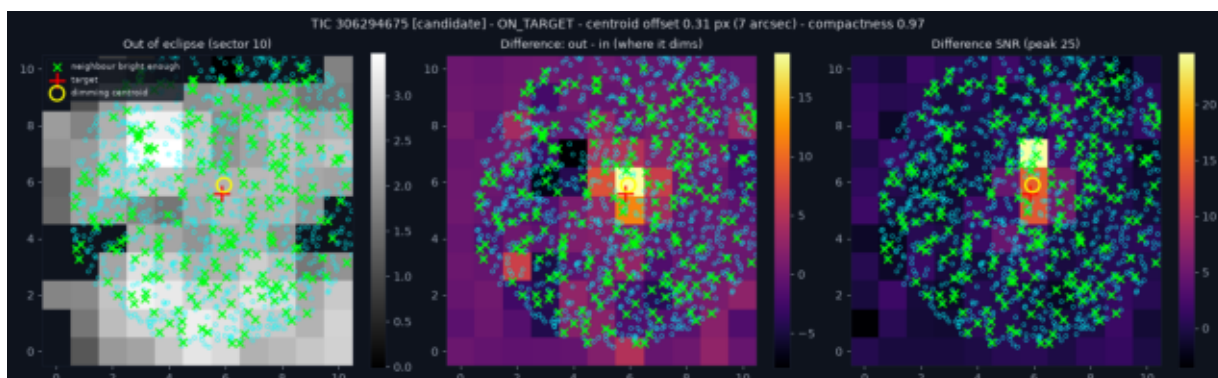
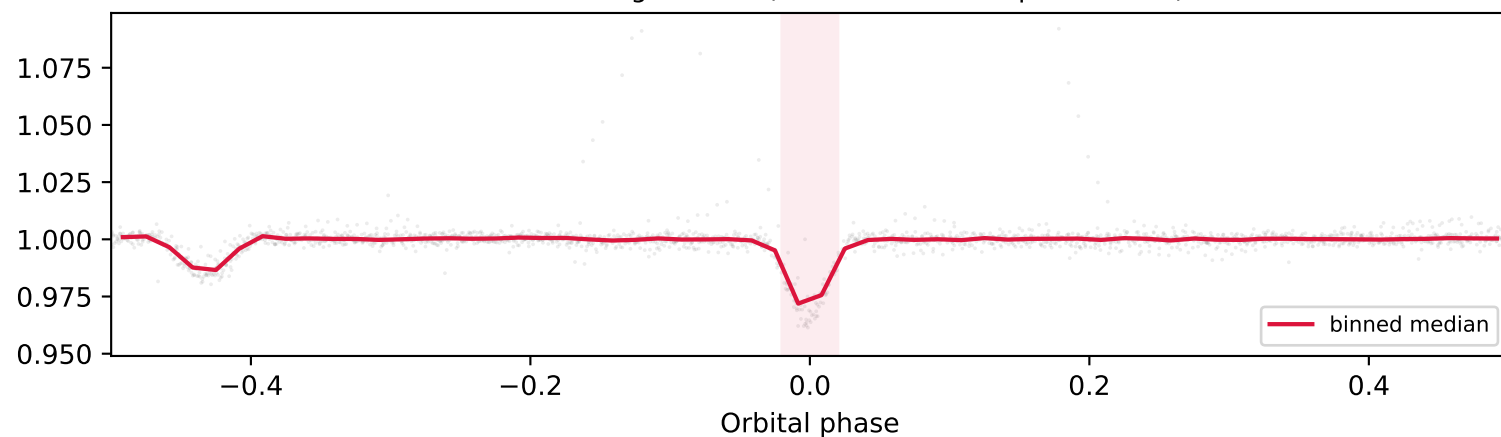
Eclipses observed 14 (gate ≥ 3)
Robust depth ratio 1.00 (gate ≥ 0.50)
Robust depth sigma 81.4 (gate ≥ 7.0)
Odd/even z-score 0.15 (consistent)
Secondary eclipse detected, phase -0.43, 5.6 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 10 light-curve dip in this sector: 86.0 sigma t0 refined by -0.0050 phase
Difference-image peak SNR 24.67 (gate ≥ 5.0) Compactness 0.97 (gate ≥ 0.50)
Dimming centroid offset 0.31 px (7 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 306294642
0.54 px away, dTmag +2.96 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



TIC 43785616

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 8.955651 d (ALIASED, see below)
-> corrected period: 17.911302 d
Epoch (t0) 2175.2083 BTJD
Duration 0.3500 d
Depth 1.146 % (11,465 ppm)
SNR 160.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

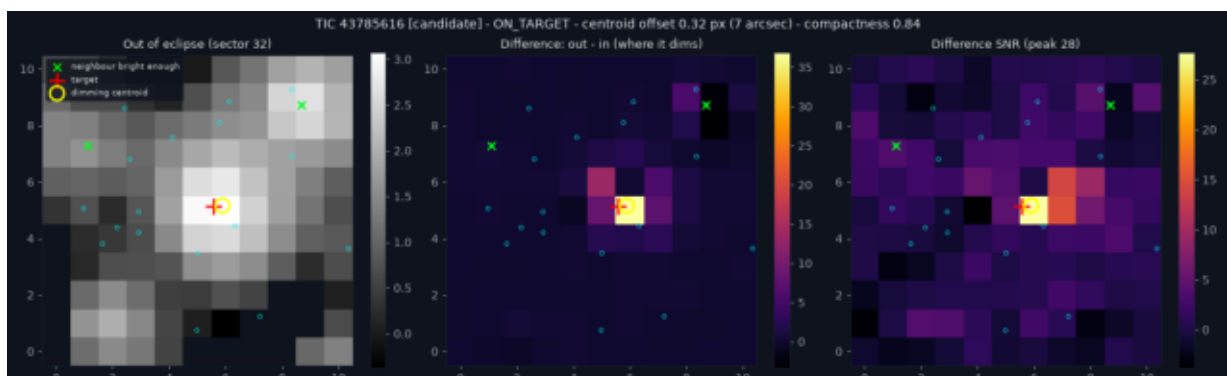
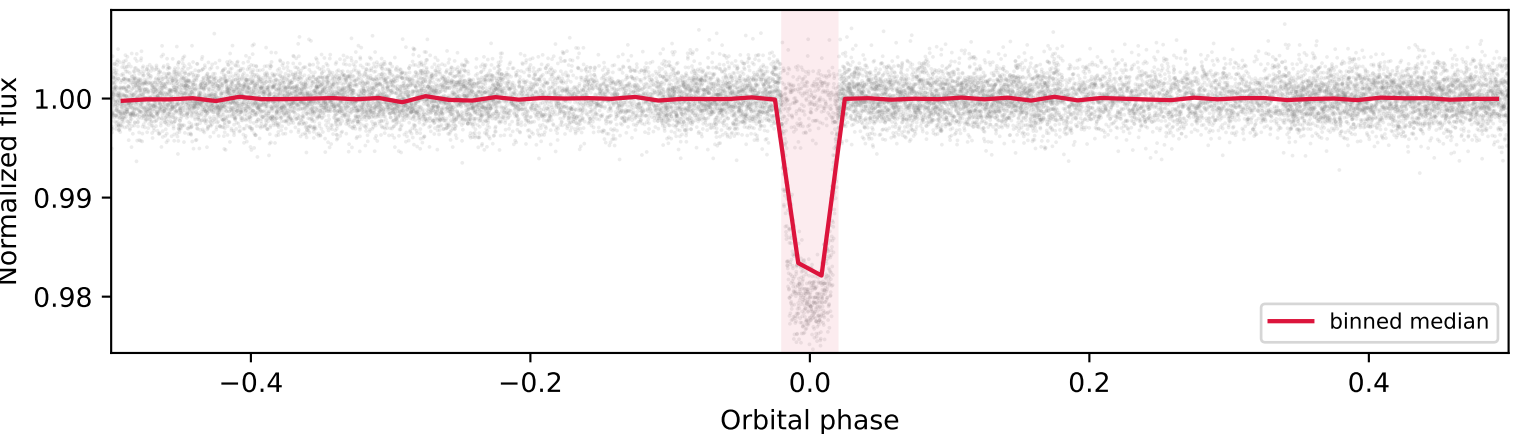
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.28 (gate ≥ 0.50)
Robust depth sigma 175.3 (gate ≥ 7.0)
Odd/even z-score 137.15 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 32 light-curve dip in this sector: 180.8 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 27.74 (gate ≥ 5.0) Compactness 0.84 (gate ≥ 0.50)
Dimming centroid offset 0.32 px (7 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 43785604
4.54 px away, dTmag +1.01 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 137.15 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 17.911302 d, in any write-up.

TIC 176348607

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 1.482622 d (ALIASED, see below)
-> corrected period: 2.965244 d
Epoch (t0) 1492.2380 BTJD
Duration 0.0580 d
Depth 8.361 % (83,611 ppm)
SNR 160.7

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

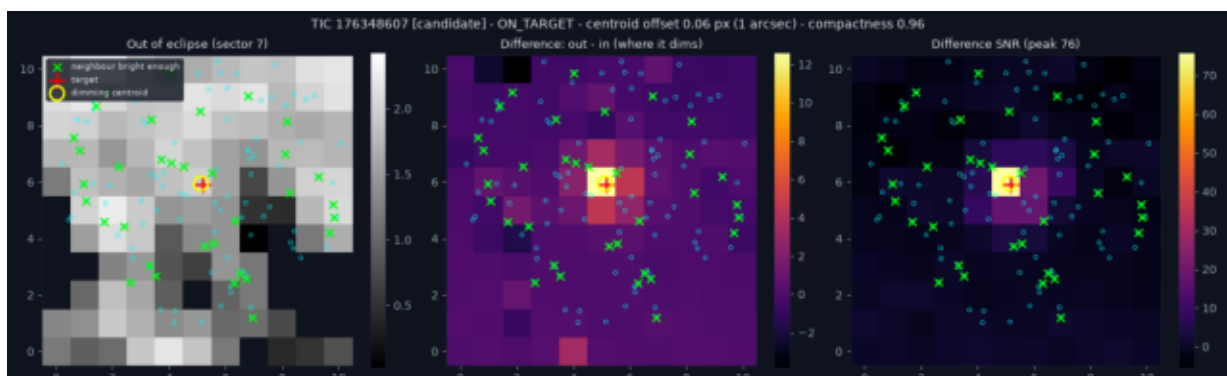
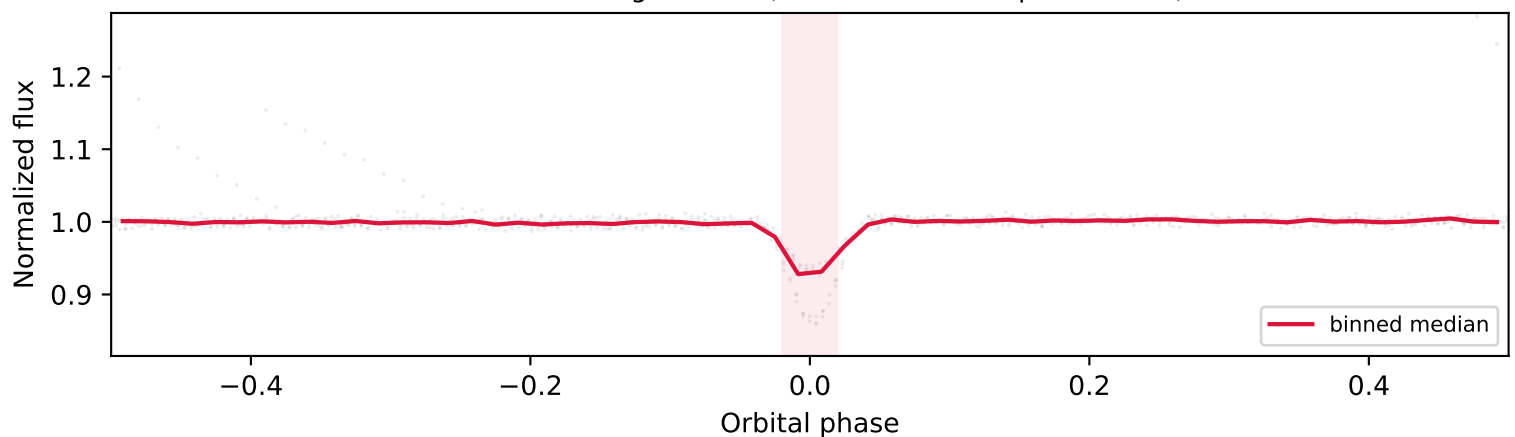
Eclipses observed 14 (gate ≥ 3)
Robust depth ratio 0.81 (gate ≥ 0.50)
Robust depth sigma 69.5 (gate ≥ 7.0)
Odd/even z-score 5.87 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 7 light-curve dip in this sector: 89.7 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 75.98 (gate ≥ 5.0) Compactness 0.96 (gate ≥ 0.50)
Dimming centroid offset 0.06 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 176348610
0.55 px away, dTmag +1.70 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 5.87 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 2.965244 d, in any write-up.

TIC 152676628

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 5.259929 d
Epoch (t0) 2968.7134 BTJD
Duration 0.1700 d
Depth 1.012 % (10,121 ppm)
SNR 159.1

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.03 (gate ≥ 0.50)
Robust depth sigma 128.1 (gate ≥ 7.0)
Odd/even z-score 1.34 (consistent)
Secondary eclipse not detected

CATALOG CROSS-MATCH (novelty)

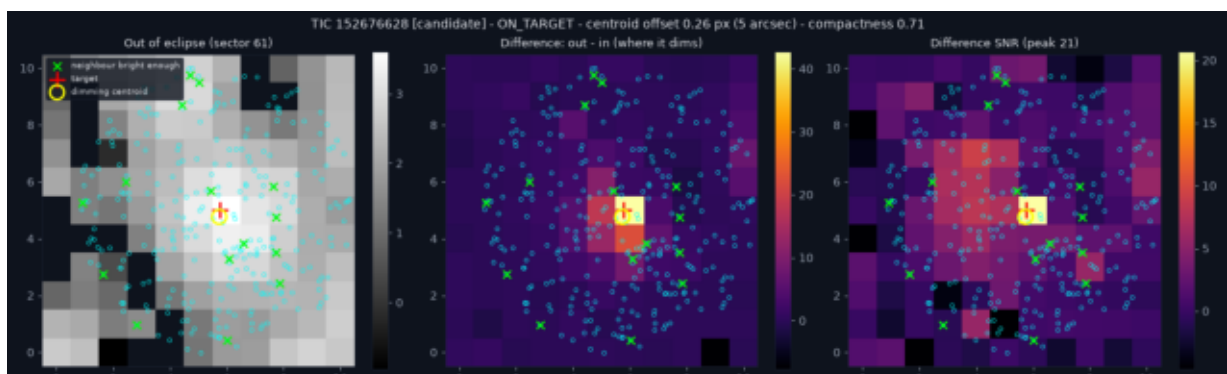
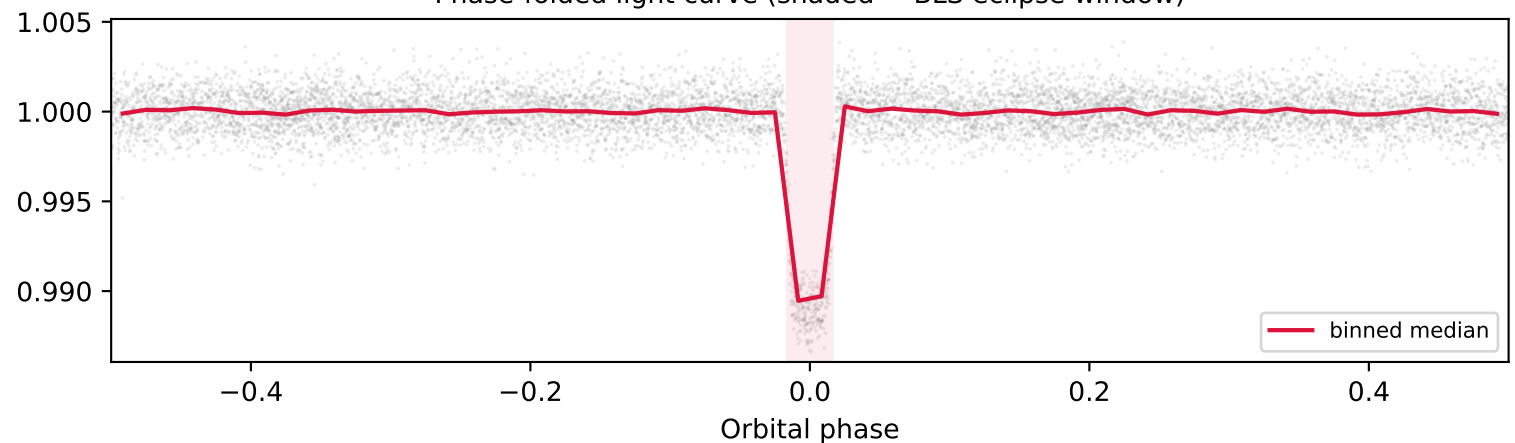
VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 61 light-curve dip in this sector: 114.8 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 20.73 (gate ≥ 5.0) Compactness 0.71 (gate ≥ 0.50)
Dimming centroid offset 0.26 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 152676613
0.97 px away, dTmag +3.93 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 434321668

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 5.563850 d (ALIASED, see below)
-> corrected period: 11.127701 d
Epoch (t0) 1473.0593 BTJD
Duration 0.1200 d
Depth 18.479 % (184,787 ppm)
SNR 151.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

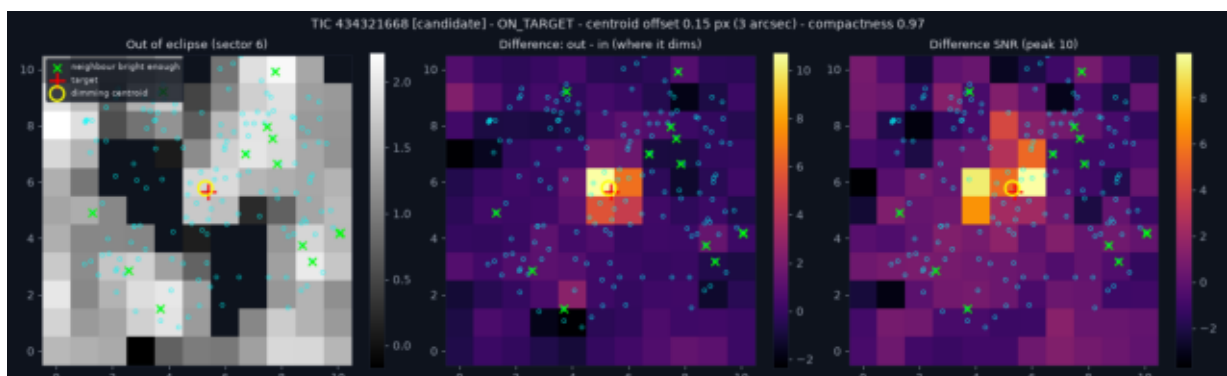
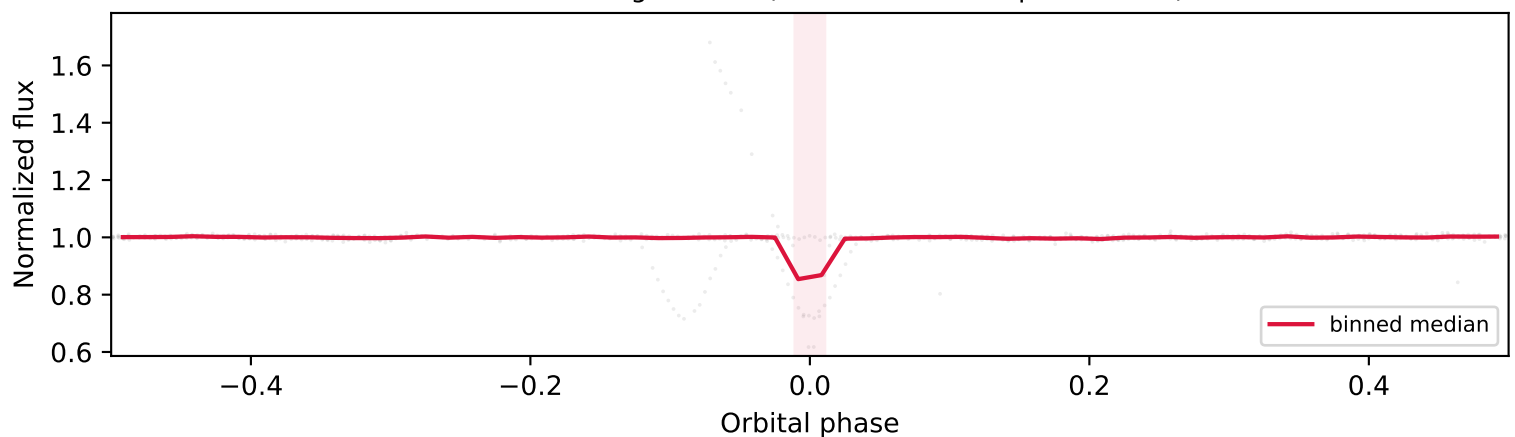
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.33 (gate ≥ 0.50)
Robust depth sigma 131.6 (gate ≥ 7.0)
Odd/even z-score 3.82 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 6 light-curve dip in this sector: 128.8 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 9.97 (gate ≥ 5.0) Compactness 0.97 (gate ≥ 0.50)
Dimming centroid offset 0.15 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 716657200
1.88 px away, dTmag +1.02 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 3.82 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 11.127701 d, in any write-up.

TIC 445266372

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 5.806270 d (ALIASED, see below)
-> corrected period: 11.612539 d
Epoch (t0) 1469.2400 BTJD
Duration 0.1200 d
Depth 8.844 % (88,441 ppm)
SNR 106.4

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

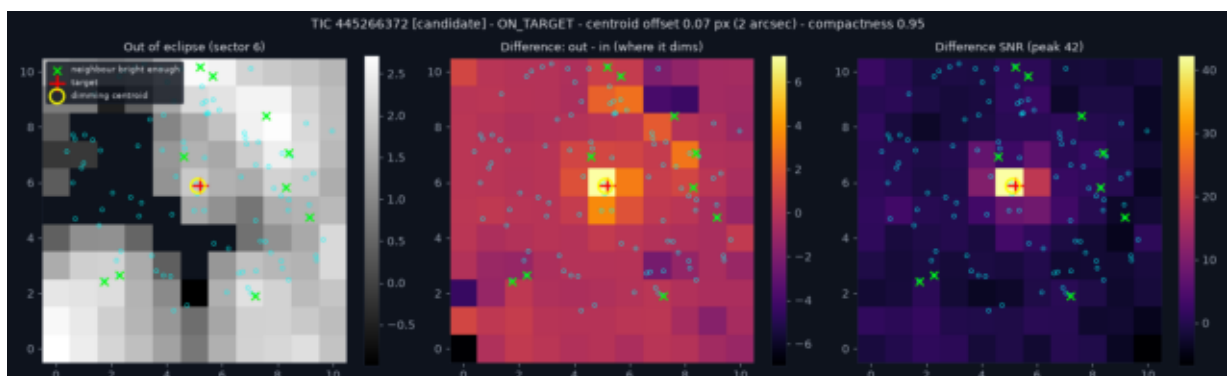
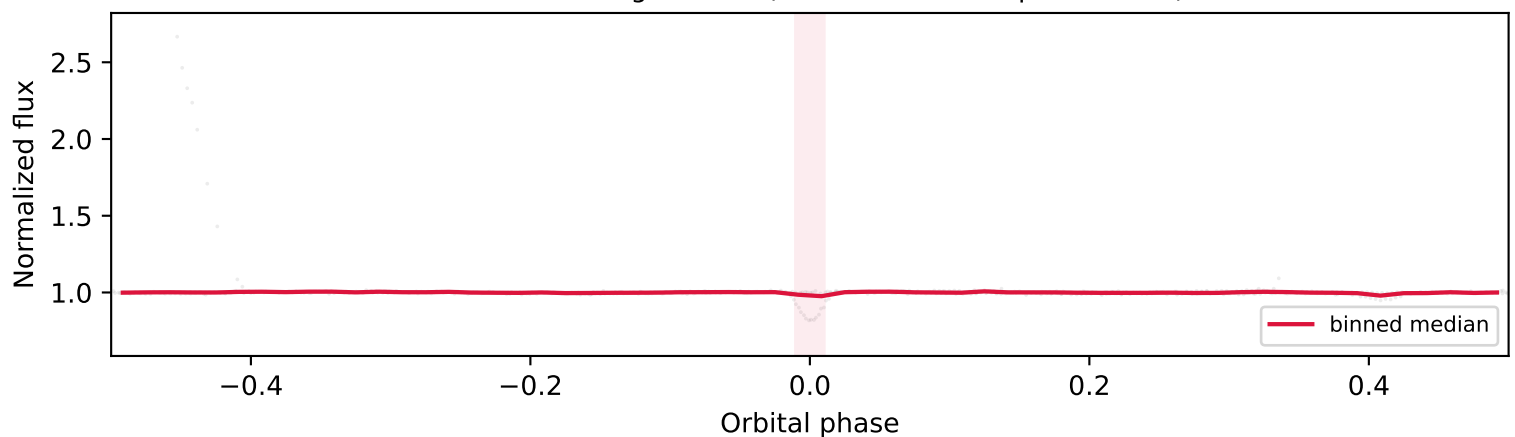
Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.11 (gate ≥ 0.50)
Robust depth sigma 65.9 (gate ≥ 7.0)
Odd/even z-score 3.62 (ALIAS FLAGGED)
Secondary eclipse detected, phase 0.41, 6.7 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 6 light-curve dip in this sector: 72.3 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 42.35 (gate ≥ 5.0) Compactness 0.95 (gate ≥ 0.50)
Dimming centroid offset 0.07 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 445267694
1.20 px away, dTmag +1.99 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 3.62 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 11.612539 d, in any write-up.

TIC 392559026

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 2.671276 d
Epoch (t0) 2308.7640 BTJD
Duration 0.0840 d
Depth 4.302 % (43,025 ppm)
SNR 90.1

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

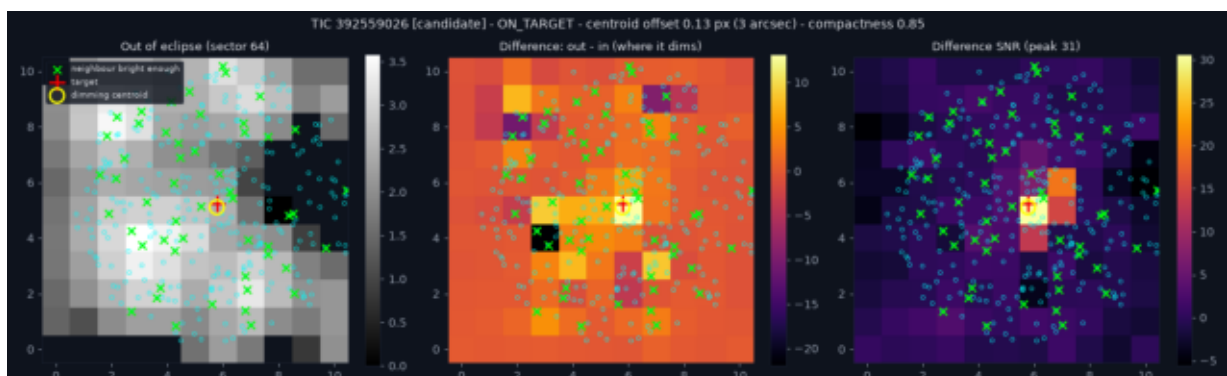
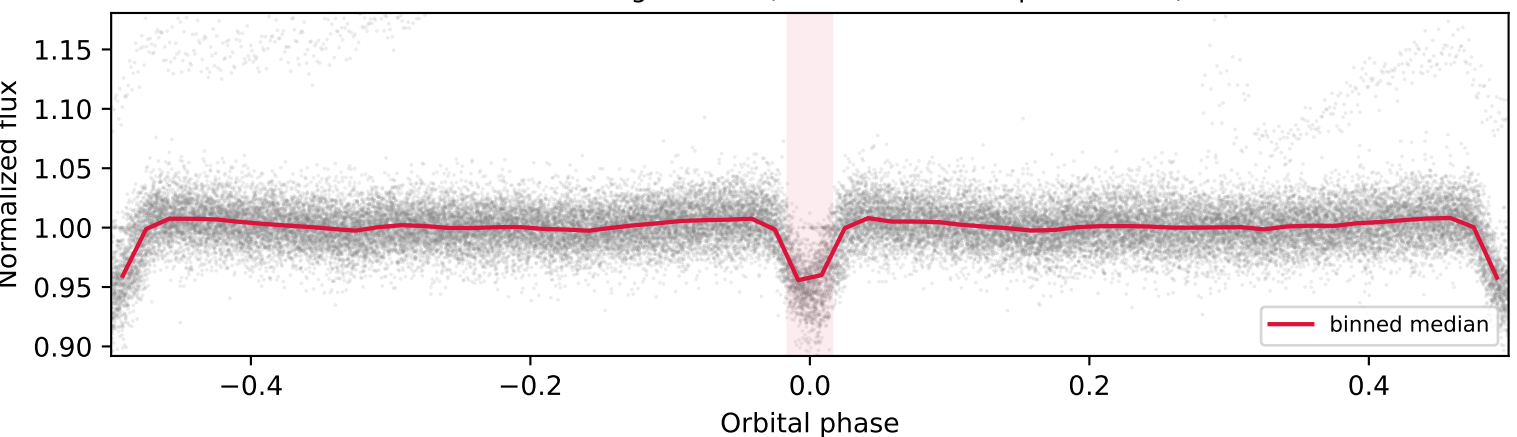
Eclipses observed 48 (gate ≥ 3)
Robust depth ratio 1.05 (gate ≥ 0.50)
Robust depth sigma 75.8 (gate ≥ 7.0)
Odd/even z-score 2.97 (consistent)
Secondary eclipse detected, phase 0.49, 5.7 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 64 light-curve dip in this sector: 38.4 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 30.60 (gate ≥ 5.0) Compactness 0.85 (gate ≥ 0.50)
Dimming centroid offset 0.13 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 392559018
0.58 px away, dTmag +1.37 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 258490713

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 4.240042 d
Epoch (t0) 1625.6508 BTJD
Duration 0.1200 d
Depth 7.295 % (72,952 ppm)
SNR 89.5

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

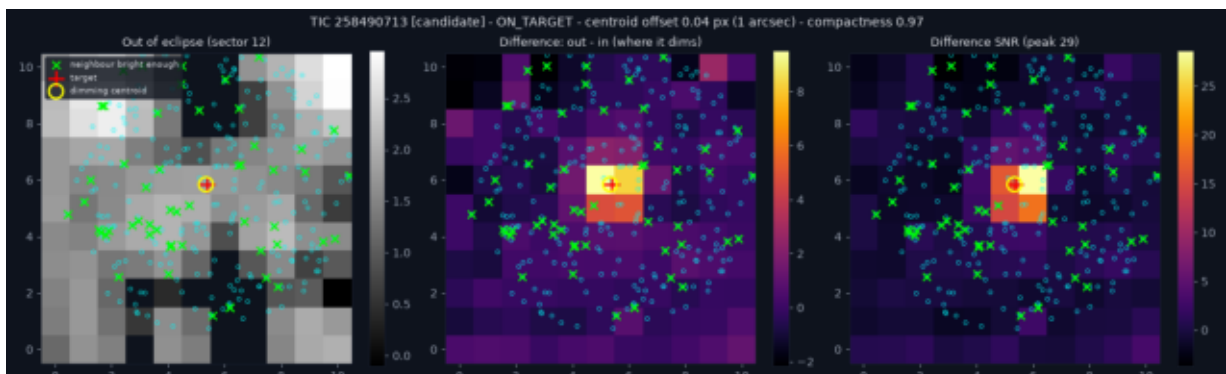
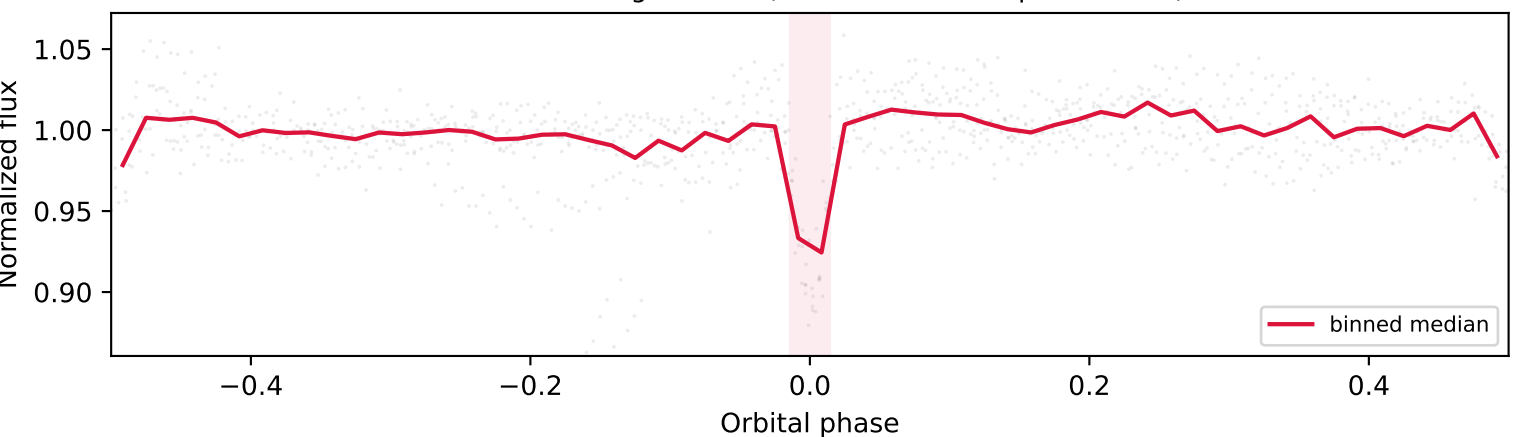
Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 0.98 (gate ≥ 0.50)
Robust depth sigma 22.0 (gate ≥ 7.0)
Odd/even z-score 0.89 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 12 light-curve dip in this sector: 25.0 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 28.69 (gate ≥ 5.0) Compactness 0.97 (gate ≥ 0.50)
Dimming centroid offset 0.04 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 258490707
0.95 px away, dTmag -0.11 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 152160223

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.333216 d
Epoch (t0) 1493.5558 BTJD
Duration 0.0840 d
Depth 4.815 % (48,148 ppm)
SNR 73.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

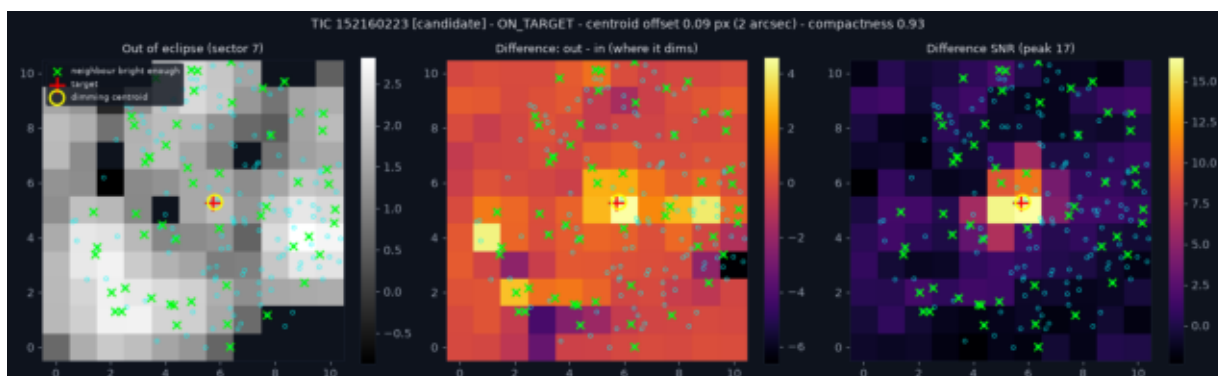
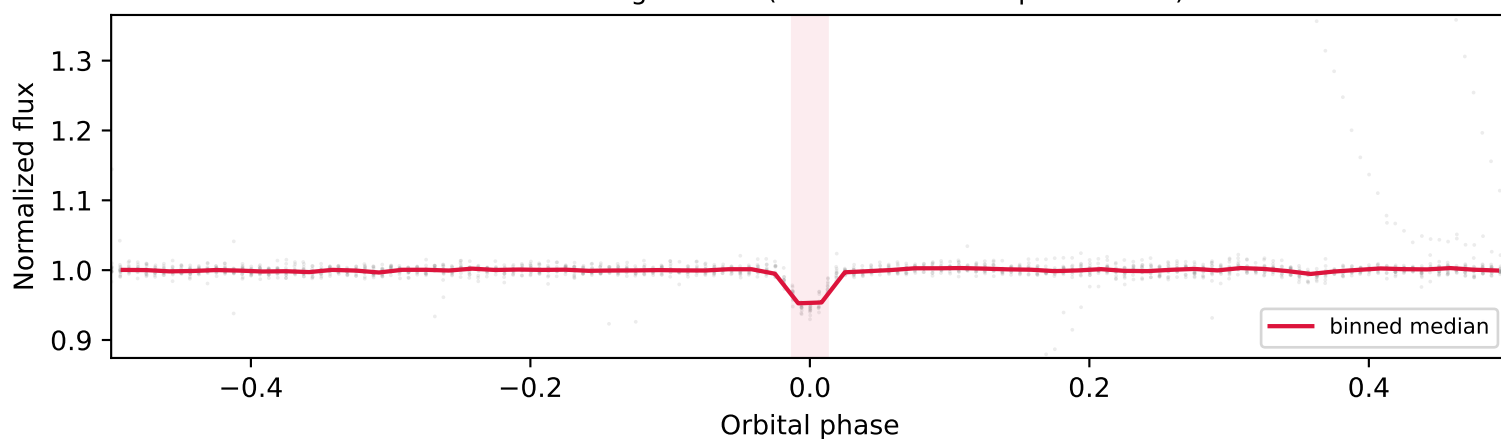
Eclipses observed 11 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 48.0 (gate ≥ 7.0)
Odd/even z-score 0.19 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 7 light-curve dip in this sector: 24.9 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 16.50 (gate ≥ 5.0) Compactness 0.93 (gate ≥ 0.50)
Dimming centroid offset 0.09 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 152160202
0.95 px away, dTmag +1.80 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 65970288

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.370534 d
Epoch (t0) 1492.9125 BTJD
Duration 0.0840 d
Depth 1.169 % (11,694 ppm)
SNR 52.8

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 16 (gate ≥ 3)
Robust depth ratio 0.97 (gate ≥ 0.50)
Robust depth sigma 36.8 (gate ≥ 7.0)
Odd/even z-score 0.18 (consistent)
Secondary eclipse not detected

CATALOG CROSS-MATCH (novelty)

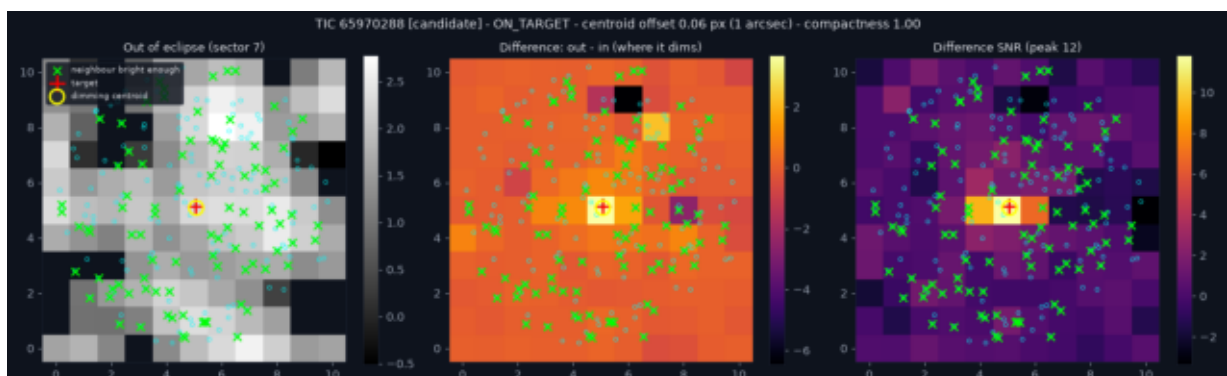
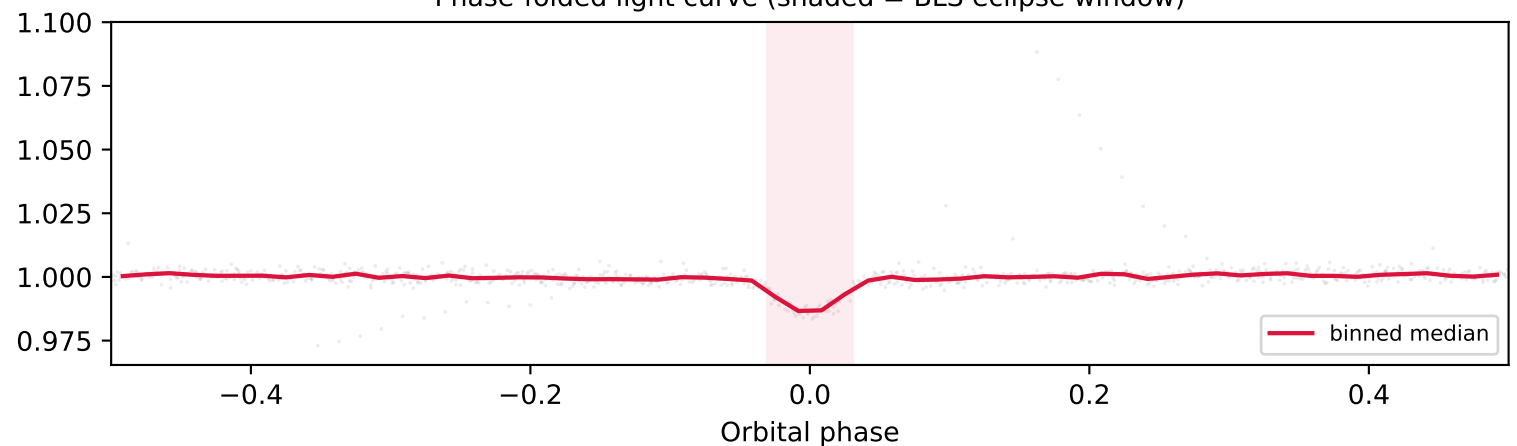
VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 7 light-curve dip in this sector: 35.9 sigma t0 refined by 0.0050 phase
Difference-image peak SNR 11.82 (gate ≥ 5.0) Compactness 1.00 (gate ≥ 0.50)
Dimming centroid offset 0.06 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 65970307
0.91 px away, dTmag +3.34 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 467962259

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 4.061582 d
Epoch (t0) 1657.6207 BTJD
Duration 0.1200 d
Depth 3.893 % (38,933 ppm)
SNR 48.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

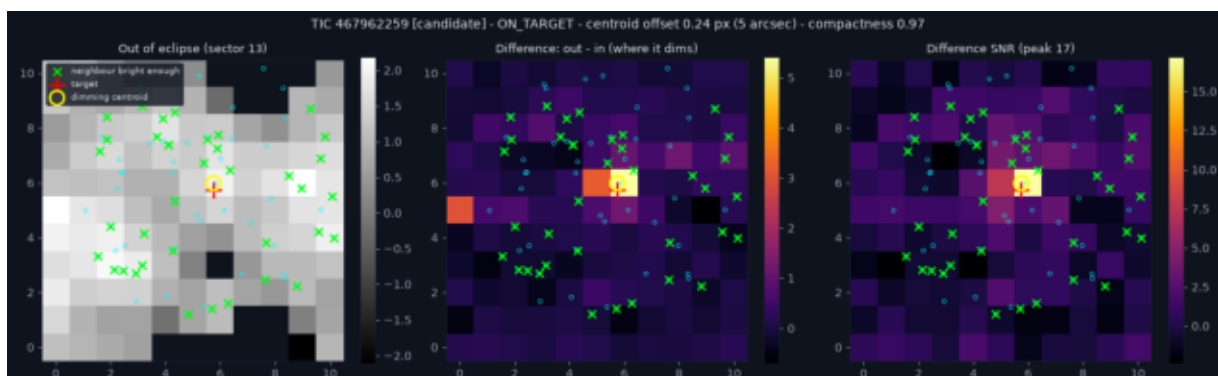
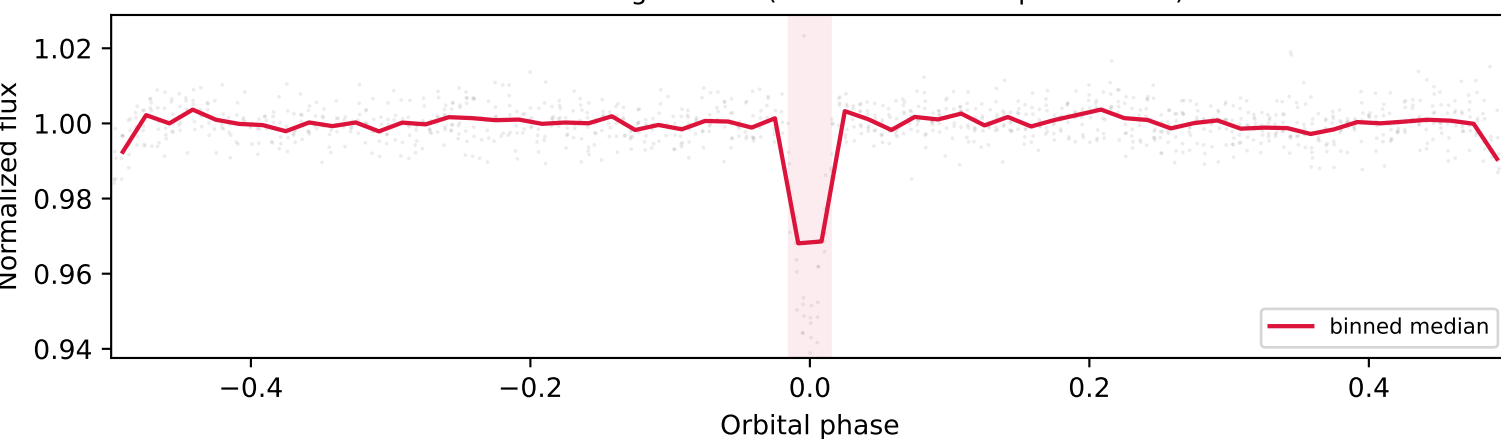
Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 1.00 (gate ≥ 0.50)
Robust depth sigma 37.9 (gate ≥ 7.0)
Odd/even z-score 0.40 (consistent)
Secondary eclipse detected, phase 0.49, 5.0 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 13 light-curve dip in this sector: 40.6 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 17.22 (gate ≥ 5.0) Compactness 0.97 (gate ≥ 0.50)
Dimming centroid offset 0.24 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 467962264
0.76 px away, dTmag +0.64 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 79792004

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 15.246229 d
Epoch (t0) 2974.4817 BTJD
Duration 0.1700 d
Depth 0.092 % (917 ppm)
SNR 12.6

CATALOG CROSS-MATCH (novelty)

VSX no match
ASAS-SN no match
Gaia eclipsing-binary table no match
Gaia variable classification no match
TESS EB catalog (Prsa+2022) no match
ExoFOP TOI list no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 3 (gate >= 3)
Robust depth ratio 0.91 (gate >= 0.50)
Robust depth sigma 9.0 (gate >= 7.0)
Odd/even z-score 1.30 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 61 light-curve dip in this sector: 5.2 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 7.99 (gate >= 5.0) Compactness 0.50 (gate >= 0.50)
Dimming centroid offset 0.63 px (13 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 79791991
0.74 px away, dTmag +5.52 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

